Raghav Arora

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Professional Summary

Computer Science graduate and Infosys System Engineer with a development focus in PL/SQL, Oracle E-Business Suite R12 and production databases on a Pfizer project. Builds AI-powered, cloud and backend applications: a serverless Generative AI service on AWS, an LLM-powered Next.js app, a LangGraph agentic RAG assistant and a deep learning (ML) chest X-ray app. Infosys Certified Generative AI Professional and Oracle Certified in OCI and AI. Seeking software developer roles in AI, Gen AI, cloud and backend engineering.

Working Experience

System Engineer **05/2026 – Present**

Infosys

- Developed and optimized multiple PL/SQL clean-up and performance-tuning scripts for production databases, for example one that cut query execution time from 2 hours to 15 minutes (an 87.5% reduction), significantly improving the performance of routine database operations
- Diagnosed and fixed production defects and resolved technical tickets as part of the L3 engineering team on a Pfizer pharmacovigilance and safety-data project, managed through ServiceNow
- Maintained and administered Oracle Argus databases, running refresh operations with APC, database and UI verification and documentation across pre- and post-refresh cycles, plus LDAP and non-LDAP access provisioning and ESDM and web-server refreshes
- Gained early exposure to AI-driven automation tooling for regulatory data workflows before moving into core database and infrastructure engineering
- Earned the Oracle Argus certification supporting enterprise pharmacovigilance and safety-data systems work

System Engineer Trainee **11/2025 – 04/2026**

Infosys

- Completed foundation training in Java, OOP and Data Structures & Algorithms, solving 50+ algorithmic problems with a focus on time and space complexity
- Worked with HSQLDB and Oracle SQL across 3 Agile sprint cycles, applying SDLC practices and improving query execution efficiency by 15%
- Built an Oracle EBS R12 Procure-to-Pay capstone: designed the data model with 3-way matching (purchase order, receipt, invoice), wrote PL/SQL procedures, triggers and collections to validate and load staging data into base tables, and automated the load with registered concurrent programs

Python Project Developer **06/2023 – 08/2023**

NetSmartz

- Developed backend and data-manipulation skills in Python and SQL by delivering 3 real-world projects
- Built and optimized 4 RESTful APIs for data retrieval between Python applications and SQL databases, reducing data transfer latency by about 20%
- Used NumPy and Pandas to process datasets of 10,000+ records, cutting manual data formatting time by 30%

Internship Trainee **06/2023 – 08/2023**

APSSDC

- Engineered a predictive model for brain stroke detection in Python, applying data analytics and machine learning across 5,000+ patient records to identify key risk factors
- Used Pandas, NumPy and Scikit-learn for analysis and model building, achieving 85% accuracy to support healthcare decision-making
- Applied data preprocessing and feature selection across 15+ medical variables, improving the model's evaluation metrics by 12%

Projects

Code Complexity Analyzer: Serverless Generative AI Code Analysis on AWS

Python, AWS Lambda, API Gateway, DynamoDB, S3, AWS SAM, Groq / Gemini / OpenAI APIs *GitHub — Live Demo*

- Built and deployed a serverless Generative AI service that uses LLMs to return Big-O time and space complexity, a bottleneck explanation and an optimized refactor for Java, Python, JavaScript and C++, with the full stack (Lambda, HTTP API, DynamoDB, S3) defined in an AWS SAM template
- Designed a pluggable LLM provider layer (abstract base class with Groq, Gemini and OpenAI backends) and a language-aware prompt builder enforcing a strict JSON schema, so models can be swapped by configuration; malformed or failed model responses return clean 502/503 errors
- Added request validation (language allow-list, 4,000-character input cap) that rejects bad payloads before any model call to keep token spend bounded
- Wrote 30 pytest unit tests covering happy paths, validation, routing and provider failures; stored analysis history in DynamoDB (SQLite locally) and built a browser UI with a CodeMirror editor, original-vs-optimized diff and a Chart.js comparison

ApplyTrail: AI-Powered Job Application Tracker & Resume Tailor

Next.js 14, TypeScript, Prisma, SQLite, NextAuth, Gmail API, OpenAI / Gemini APIs *GitHub*

- Built a full-stack application (about 9,000 lines of TypeScript) that connects to Gmail over OAuth, filters recruiter and ATS emails (Greenhouse, Lever, Workday, Ashby, LinkedIn) and classifies each as applied, reply, interview, rejection or offer using an LLM with a strict JSON schema and confidence threshold, falling back to a rule-based classifier when no API key is set
- Engineered an incremental sync pipeline that processes emails oldest to newest in batches of 200, runs metadata and AI calls in parallel, uses a per-user lock to prevent overlapping syncs, and saves each result with its processed-email marker in a single Prisma transaction so failures retry safely without duplicates
- Built a resume-tailoring studio that parses PDF, DOCX, TXT and Markdown resumes, scrapes job-description URLs, extracts JD keywords and rewrites bullets in three alignment modes, with word-level diffs, matched and missing keyword scores, per-edit toggles, revision history and export
- Added a dashboard with search, filters, sorting, pagination, CSV export, analytics charts, automatic Gmail labelling and 6-hour auto-sync on a local-first SQLite database

HealthLens: Chest X-Ray Disease Detection Web App (Deep Learning)

React, Tailwind CSS, Firebase, FastAPI, PyTorch, Captum *GitHub*

- Built a full-stack machine learning (ML) web application around a PyTorch ResNet-18 classifier that detects pneumonia and COVID-19 from chest X-rays, serving real-time inference through a FastAPI backend
- Integrated Captum's Occlusion algorithm for post hoc explainability, generating saliency maps that show which image regions drove each prediction, for clinical review
- Developed the React and Tailwind CSS frontend with Firebase authentication, letting users upload chest X-ray images and receive detailed diagnostic reports

Agentic Research Assistant: Generative AI RAG Agent for arXiv

Python, LangGraph, LangChain, Gemini, ChromaDB, PyMuPDF, Gradio *GitHub*

- Built a Generative AI agent with LangGraph (ReAct) on Gemini that decides per query whether to retrieve from ingested papers, search the web or ask a clarifying question, with tool calling, checkpointed conversation state, streaming responses and a 10-iteration safety cap
- Built the ingestion pipeline (arXiv fetch, PDF parsing with pymupdf4llm, overlapping chunking with paper metadata, embedding and storage in ChromaDB) using deterministic chunk IDs (arXiv ID plus chunk index) so re-ingestion is idempotent
- Required cited answers (title, authors, arXiv ID) and instructed the agent to flag low-similarity retrievals instead of answering confidently; delivered through a Gradio UI with per-topic collections and live ingestion progress

Skills

- **Languages:** Python, Java, TypeScript, JavaScript, PL/SQL, SQL, C/C++
- **AI, ML & Gen AI:** Generative AI, LLMs, Agentic AI, RAG, LangGraph, LangChain, Prompt Engineering, ChromaDB, LLM APIs (Gemini, OpenAI, Groq), Machine Learning, Deep Learning, PyTorch, Captum, Scikit-learn, Pandas, NumPy, CNNs
- **Backend & Web:** FastAPI, Flask, REST APIs, Next.js, React, Tailwind CSS, Prisma, NextAuth, Gmail API, Firebase, HTML, CSS
- **Cloud & Tools:** AWS (Lambda, API Gateway, DynamoDB, S3, SAM), Oracle Cloud Infrastructure, Git, GitHub, Unix, pytest, ServiceNow
- **Databases & Enterprise:** Oracle SQL, PL/SQL, SQLite, DynamoDB, HSQLDB, Oracle E-Business Suite R12, Oracle Argus, ERP
- **Concepts:** Data Structures & Algorithms, Agile/SDLC, Cloud Computing
- **Spoken Languages:** English (Professional), Hindi (Native)

Certifications

- **Infosys Certified Generative AI Professional Foundation**, Infosys, 09/2026
- **Oracle Cloud Infrastructure Certified Foundations Associate**, Oracle University, 07/2026 (ID 330061829OCI26FNDCFA)
- **Oracle Cloud Infrastructure Certified AI Foundations Associate**, Oracle University, 07/2026 (ID 330061829OCI26AICFA)
- **Oracle AI Database Certified Foundations Associate**, Oracle University, 07/2026 (ID 330061829OCI26DCFA)
- **Oracle E-Business Suite (EBS) R12 Certification**, Infosys, 04/2026
- **Oracle Argus Certification**, Infosys, 04/2026

Education

B.Tech, Computer Science Engineering **CGPA: 7.32**

SRM University AP *2021 – 2025*

Senior Secondary (CBSE) **Grade: 90.0%**

Delhi Public School Indirapuram *2020 – 2021*